```
import socket Untitled-1 2 ●   u1.py
C: > Users > HDC0222008 > Downloads > Packet Sniffer > u1.py > ...
1   import socket
2   import struct
3   sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
4   host = socket.gethostbyname(socket.gethostname())
5   sniffer.bind((host, 0))
6   sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
7   sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)
8   print("Packet Sniffer Started...\n")
9   try:
10      while True:
11          raw_data, addr = sniffer.recvfrom(65535)
12          ip_header = raw_data[0:20]
13          iph = struct.unpack('!BBHHHBBH4s4s', ip_header)
14          version = iph[0] >> 4
15          ihl = iph[0] & 0xf
16          ttl = iph[5]
17          protocol = iph[6]
18          source = socket.inet_ntoa(iph[8])
19          destination = socket.inet_ntoa(iph[9])
20          print("----------------------------------------")
21          print("Version      :", version)
22          print("Header Length:", ihl * 4)
23          print("TTL          :", ttl)
24          print("Protocol     :", protocol)
25          print("Source IP    :", source)
26          print("Destination  :", destination)
27          print("Packet Length:", len(raw_data))
28  except KeyboardInterrupt:
29      sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
30      print("\nStopped")
```

```
----------------------------------------
Version      : 4
Header Length: 20
TTL          : 1
Protocol     : 17
Source IP    : 172.16.8.73
Destination  : 224.0.0.251
Packet Length: 371
----------------------------------------
Version      : 4
Header Length: 20
TTL          : 1
Protocol     : 17
Source IP    : 172.16.8.73
Destination  : 224.0.0.251
Packet Length: 307
----------------------------------------
Version      : 4
Header Length: 20
TTL          : 1
Protocol     : 17
Source IP    : 172.16.8.139
Destination  : 239.255.255.250
Packet Length: 203
----------------------------------------
Version      : 4
Header Length: 20
TTL          : 1
Protocol     : 17
Source IP    : 172.16.8.144
Destination  : 224.0.0.251
Packet Length: 172
----------------------------------------
Version      : 4
Header Length: 20
TTL          : 1
Protocol     : 17
Source IP    : 172.16.8.7
Destination  : 224.0.0.251
Packet Length: 172
```

4. PACKET SNIFFING USING RAW SOCKETS IN PYTHON

AIM: To implement a packet sniffer using raw sockets in Python that captures live network traffic & parses Ethernet, IP, transport layer (TCP/UDP/ICMP) headers.

PROCEDURE:

Step 1: Set up the Lab Environment

Step 2: Understand Packet structure you'll Pharse

Step 3: Create the Raw Socket

Step 4: Parse the Ethernet Header

Step-5: Parse the IP Header

Step-6: Parse TCP/UDP/ICMP Headers

Step 7: Write the main sniffing loop

Step 8: Run the Sniffer

Step 9: Extend the Lab

Step 10: Lab Report - What to Document

RESULT: Thus the implementation of Packet sniffing using Raw Sockets in Python was successfully completed.